

A Diagnostic Analyses

This appendix provides additional diagnostic analyses that support the regime-based interpretation discussed in the main paper. We analyze the interaction between forecasting and alignment objectives through gradient consistency and evaluate robustness under noisy numerical inputs.

A.1 Gradient Consistency

To examine whether the alignment objectives cooperate with the forecasting optimization, we measure the *gradient consistency* (GC) between the prediction loss and an alignment loss using the cosine similarity between their gradients:

$$\text{GC} = \cos(\nabla_{\theta}\mathcal{L}_{\text{pred}}, \nabla_{\theta}\mathcal{L}_{\text{align}}) = \frac{\langle \nabla_{\theta}\mathcal{L}_{\text{pred}}, \nabla_{\theta}\mathcal{L}_{\text{align}} \rangle}{\|\nabla_{\theta}\mathcal{L}_{\text{pred}}\| \|\nabla_{\theta}\mathcal{L}_{\text{align}}\|}. \quad (29)$$

Here θ denotes the trainable parameters of the model. $\mathcal{L}_{\text{pred}}$ is the forecasting objective (MSE), and $\mathcal{L}_{\text{align}}$ represents an alignment loss (SA, MA, or DA). Positive GC values indicate cooperative optimization directions, while negative values indicate gradient conflict.

We compute GC per parameter tensor and visualize the resulting distributions as histograms.

SA-dominant case. On TCAP-FM, the GC distribution of SA-only remains relatively compact and mildly positive, indicating stable cooperation between semantic alignment and forecasting optimization. In contrast, MA-only exhibits a broader distribution with higher variance, suggesting less consistent optimization behavior.

MA-dominant case. On TCAP-WN, MA-only shows stronger but higher-variance interactions with the forecasting objective, indicating that movement alignment can induce larger updates while occasionally conflicting with forecasting gradients. SA-only remains comparatively weak in this setting.

DA-dominant case. On TMMD-SG (Time-MMD SocialGood), dual alignment produces a more compact and positively shifted GC distribution compared with MA-only, suggesting that combining SA and MA stabilizes alignment updates and improves cooperation with forecasting optimization.

Figure 2 visualizes representative GC distributions for these regimes.

A.2 Noise Injection Robustness

To evaluate robustness under corrupted numerical observations, we inject Gaussian noise into the numerical inputs:

$$\mathbf{X}^{(\eta)} = \mathbf{X} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}\left(\mathbf{0}, (\eta \cdot \text{std}(\mathbf{X}))^2\right), \quad (30)$$

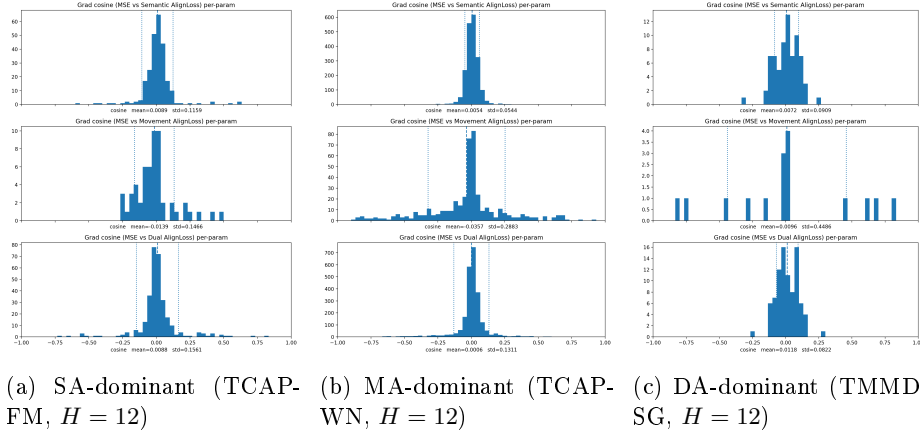


Fig. 2: Per-parameter gradient cosine similarity (GC) between forecasting loss (MSE) and alignment losses. Each panel reports three variants (top: SA-only; middle: MA-only; bottom: DA).

where η controls the noise level and $\text{std}(\cdot)$ is computed along the time dimension.

The forecasting error under noise is evaluated as

$$\text{MSE}(\eta) = \mathbb{E} \left[\left\| \hat{\mathbf{Y}}(\mathbf{X}^{(\eta)}, \mathbf{S}) - \mathbf{Y} \right\|_2^2 \right]. \quad (31)$$

As η increases, forecasting performance gradually degrades for all methods, but the degradation rates vary across alignment designs.

We observe three representative robustness regimes:

- **DA-dominant robustness.** On TMMD-C (Time-MMD Climate), dual alignment consistently achieves the lowest MSE across noise levels, suggesting that combining semantic and movement constraints helps preserve predictive stability.
- **SA-dominant robustness.** On TMMD-SG, SA-only remains the most robust across noise levels, indicating that semantic regularization alone can effectively stabilize text-conditioned representations.
- **MA-dominant robustness.** On NF-FA (NewsForecast-Finance-Australia), movement alignment is the most resilient to noise perturbations. Interestingly, MA-dominant robustness mainly appears in multivariate forecasting settings, suggesting that preserving dynamic structural patterns becomes particularly beneficial when multiple variables interact.

Overall, these results indicate that alignment mechanisms improve robustness through complementary mechanisms: semantic alignment stabilizes cross-modal semantics, movement alignment preserves dynamic structure, and their combination can further enhance robustness in certain regimes.

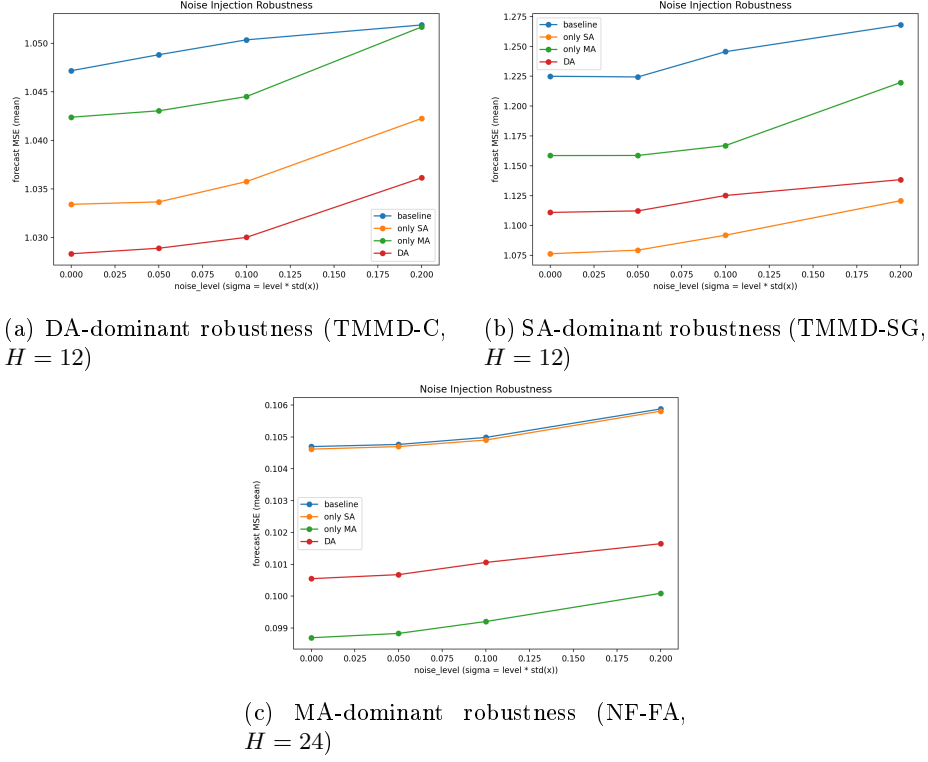


Fig. 3: Noise injection robustness under Gaussian perturbations with $\sigma = \eta \cdot \text{std}(\mathbf{X})$.

B Additional Transfer Results

To further examine whether the proposed dual alignment (DA) mechanism can generalize beyond the DA-Transformer architecture, we transplant the DA objectives and modules onto several representative forecasting backbones. The resulting models are denoted with the suffix `_DA`.

Table 3 reports the detailed dataset-level results comparing each backbone with its DA-enhanced counterpart. In the table, bold indicates the overall best result (SOTA) across all methods, while underline indicates the second-best result. For each DA variant, the symbols $\uparrow$ and $\downarrow$ indicate whether the corresponding metric improves or degrades relative to its original backbone.

The results show that DA-enhanced models frequently outperform their original counterparts across multiple datasets and architectures. Although improvements are not uniform across all settings, DA variants achieve new SOTA results on several benchmarks, suggesting that the proposed alignment mechanism is not tied to a specific architecture and can serve as a generally applicable design component for multimodal time-series forecasting models.

B.1 Transfer Metrics Details

This appendix reports the detailed statistics used to evaluate the transfer effectiveness of dual alignment across different backbone architectures.

The following metrics are used:

- **All-metric Win Rate (AWR)**: percentage of datasets where both MSE and MAE improve after introducing DA.
- **Partial-metric Win Rate (PWR)**: percentage of datasets where at least one metric improves.
- **Mean Relative Improvement (MRI)**: average relative improvement over improved metrics.
- **Mean Relative Degradation (MRD)**: average relative degradation over degraded metrics.

The aggregated statistics across forecasting horizons $H = 12$ and $H = 24$ are summarized in Table 4.

Table 1: Main results (MSE/MAE) on benchmarks (Time-MMD, NewsForecast, TimeCAP, and TTC).

Data-set	Metri.	DA-Trans.	MM-T. (iTrans.)	MM-T. (Patch.)	NLin.	Ma Trans.	Uni.	Time-LLM
TMMD-H	MSE	1.0829	1.2430	<u>1.1842</u>	1.2869	1.2000	1.9075	1.6391
	MAE	<u>0.6747</u>	0.6653	0.7005	0.8114	0.7234	0.9025	0.8673
TMMD-A	MSE	0.4329	0.4299	0.4513	0.4854	1.1749	1.0529	0.5412
	MAE	0.4554	0.4718	0.4900	<u>0.4686</u>	0.7643	0.7848	0.5930
TMMD-C	MSE	1.0053	1.1604	<u>1.0239</u>	1.0645	1.2241	1.2081	1.1624
	MAE	0.8096	0.8984	<u>0.8329</u>	0.8239	0.8815	0.8782	0.8646
TMMD-Enc	MSE	0.0145	<u>0.0164</u>	0.0181	0.0448	1.1295	5.9978	0.0405
	MAE	0.0952	<u>0.1035</u>	0.1086	0.1685	1.0094	2.4338	0.1622
TMMD-En	MSE	0.1588	0.1825	<u>0.1644</u>	0.1654	0.2985	1.0659	0.2568
	MAE	0.2898	0.3049	<u>0.2943</u>	0.2979	0.4101	0.8020	0.3758
TMMD-Env	MSE	0.3988	<u>0.4042</u>	0.4124	0.4117	0.4310	0.5555	0.4988
	MAE	0.4612	0.4646	<u>0.4621</u>	0.4685	0.4876	0.5915	0.5072
TMMD-S	MSE	<u>99.4314</u>	101.6838	100.6717	94.9721	112.9370	114.6854	100.6729
	MAE	<u>5.2426</u>	5.2649	5.1875	4.5411	6.3201	6.4580	5.3066
TMMD-SG	MSE	<u>1.2297</u>	1.2973	1.4221	1.3880	0.8662	1.7966	1.7402
	MAE	0.6408	0.5762	0.7207	0.8353	<u>0.6282</u>	1.1014	0.8355
TMMD-T	MSE	0.1988	0.2291	<u>0.2036</u>	0.2651	0.2857	1.4763	0.3202
	MAE	<u>0.2481</u>	0.2524	0.2321	0.3754	0.4331	1.1122	0.3787
NF-FA	MSE	0.0727	<u>0.1118</u>	0.1134	0.1326	2.1544	3.6883	0.1572
	MAE	0.1086	0.2198	0.2226	0.2537	1.1270	1.5509	<u>0.2183</u>
NF-FC	MSE	0.6753	0.6565	<u>0.6300</u>	0.5681	0.6480	0.6785	<u>0.6292</u>
	MAE	0.5614	0.5562	<u>0.5521</u>	0.5355	0.5702	0.6054	<u>0.5457</u>
NF-T	MSE	<u>1.2264</u>	1.2313	<u>1.2250</u>	1.2301	0.6257	1.3301	1.3463
	MAE	<u>0.7675</u>	0.7787	<u>0.7738</u>	0.7914	0.4410	0.8196	0.8328
TCAP-WH	MSE	0.8150	0.8112	<u>0.7964</u>	0.7793	0.8941	0.9696	0.8363
	MAE	0.6662	0.6630	<u>0.6571</u>	0.6488	0.7083	0.7696	0.6763
TCAP-WN	MSE	0.8134	0.7508	<u>0.7363</u>	0.7289	0.8871	0.9759	0.7732
	MAE	0.6884	<u>0.6659</u>	<u>0.6612</u>	0.6605	0.7442	0.7982	0.6834
TCAP-WS	MSE	0.7884	<u>0.8006</u>	0.8014	0.8084	1.1199	1.3613	0.8923
	MAE	0.6548	<u>0.6542</u>	0.6522	0.6562	0.7588	0.8638	0.6941
TCAP-FM	MSE	0.1008	0.1579	<u>0.1453</u>	0.1567	2.6243	4.4568	0.2115
	MAE	0.2342	0.2938	<u>0.2826</u>	0.2892	1.1054	1.4979	0.3404
TCAP-HM	MSE	0.3887	0.4226	0.4240	0.5744	0.7633	0.5743	0.7331
	MAE	0.4573	<u>0.4622</u>	0.4680	0.5797	0.7639	0.5792	0.6597
TCAP-HP	MSE	<u>3.5497</u>	3.7627	3.4236	4.0059	6.5429	6.3504	4.4979
	MAE	<u>0.9107</u>	<u>0.8935</u>	0.8391	0.9988	1.3960	1.4169	1.0353
TTC-C	MSE	0.6349	0.6483	<u>0.6291</u>	0.6359	0.5901	0.7735	0.6791
	MAE	0.5871	<u>0.5801</u>	<u>0.5669</u>	0.5844	0.5642	0.6758	0.5968
TTC-M7679	MSE	1.5711	2.0028	<u>1.6166</u>	2.0776	4.7009	4.8152	2.4726
	MAE	<u>0.8631</u>	0.9633	0.8590	0.9612	1.5874	1.7428	1.0200
TTC-M8355	MSE	3.0221	3.4278	<u>3.1111</u>	3.1214	3.2655	3.2098	3.5072
	MAE	1.2955	1.4224	1.3358	1.3374	1.3630	<u>1.3245</u>	1.4288
TTC-M15432	MSE	1.4930	1.6077	<u>1.5335</u>	1.6863	2.9719	3.8612	1.7142
	MAE	0.8503	0.9446	<u>0.9203</u>	1.0290	1.3949	1.5907	1.0302
TTC-M20415	MSE	3.8226	4.1907	<u>3.9465</u>	4.0105	4.6530	4.4719	4.1253
	MAE	<u>1.1376</u>	1.1986	1.1328	1.1578	1.3674	1.4090	1.2062
TTC-M24501	MSE	<u>1.0277</u>	1.2793	1.3914	1.0250	1.6490	2.5287	1.9119
	MAE	0.7126	0.7746	0.8426	<u>0.7127</u>	1.0592	1.3292	1.0412

Table 2: Ablation study (MSE / MAE) of DA-Transformer on benchmarks.

Data-set	Metr.	DA	SA only	MA only	w/o Align.
TMMD	MSE	1.0829	1.1535	<u>1.1118</u>	1.2451
-H	MAE	0.6747	0.6965	<u>0.6852</u>	0.6958
TMMD	MSE	0.4329	0.4285	0.4302	<u>0.4294</u>
-A	MAE	0.4554	<u>0.4591</u>	0.4603	0.4601
TMMD	MSE	<u>1.0053</u>	1.0072	1.0025	1.0189
-C	MAE	0.8096	<u>0.8115</u>	0.8116	0.8185
TMMD	MSE	0.0145	<u>0.0150</u>	0.0155	0.0153
-Ec	MAE	0.0952	<u>0.0980</u>	0.0993	0.0986
TMMD	MSE	<u>0.1588</u>	0.1609	0.1851	0.1584
-En	MAE	0.2898	<u>0.2902</u>	0.3123	0.2903
TMMD	MSE	<u>0.3988</u>	0.3982	0.4001	0.3990
-Env	MAE	0.4612	0.4588	0.4616	<u>0.4607</u>
TMMD	MSE	99.4314	101.7102	<u>100.1057</u>	101.3298
-S	MAE	5.2426	5.3982	<u>5.2438</u>	5.3976
TMMD	MSE	1.2297	1.2741	<u>1.2481</u>	1.2761
-SG	MAE	<u>0.6408</u>	0.6404	0.6544	0.6532
TMMD	MSE	<u>0.1988</u>	0.1993	0.1983	0.1993
-T	MAE	<u>0.2481</u>	0.2463	0.2505	0.2495
NF-FA	MSE	0.0727	0.0794	<u>0.0757</u>	0.0802
	MAE	0.1086	0.1191	<u>0.1135</u>	0.1206
NF-FC	MSE	<u>0.6753</u>	0.6807	0.6737	0.6818
	MAE	0.5614	0.5637	<u>0.5618</u>	0.5653
NF-T	MSE	1.2264	1.2351	<u>1.2278</u>	<u>1.2278</u>
	MAE	0.7675	0.7752	<u>0.7736</u>	0.7759
TCAP	MSE	0.8150	0.8206	<u>0.8152</u>	0.8161
-WH	MAE	0.6662	0.6664	0.6651	<u>0.6657</u>
TCAP	MSE	0.8134	<u>0.8077</u>	0.8014	0.8187
-WN	MAE	0.6884	<u>0.6859</u>	0.6850	0.6918
TCAP	MSE	<u>0.7884</u>	0.7980	0.7858	0.7904
-WS	MAE	0.6548	0.6548	0.6490	<u>0.6527</u>
TCAP	MSE	0.1008	0.0997	<u>0.1003</u>	0.1012
-FM	MAE	0.2342	0.2332	<u>0.2337</u>	0.2354
TCAP	MSE	0.3887	0.4016	0.3614	<u>0.3762</u>
-HM	MAE	0.4573	0.4701	0.4336	<u>0.4536</u>
TCAP	MSE	3.5497	3.5379	<u>3.5481</u>	3.6391
-HP	MAE	0.9107	<u>0.8868</u>	<u>0.8659</u>	0.8461
TTC-C	MSE	<u>0.6349</u>	0.6416	0.6310	0.6361
	MAE	0.5871	0.5873	0.5802	<u>0.5826</u>
TTC-M	MSE	1.5711	1.2726	1.4545	<u>1.4057</u>
7679	MAE	0.8631	0.7864	0.8322	<u>0.8182</u>
TTC-M	MSE	3.0221	3.1588	3.1196	<u>3.0364</u>
8355	MAE	1.2955	1.3300	1.3202	<u>1.3003</u>
TTC-M	MSE	1.4930	1.5046	1.4968	<u>1.4965</u>
15432	MAE	0.8503	0.8532	<u>0.8528</u>	0.8548
TTC-M	MSE	3.8226	3.8778	3.8214	<u>3.8220</u>
20415	MAE	1.1376	1.1445	<u>1.1377</u>	1.1391
TTC-M	MSE	1.0277	1.0309	<u>1.0302</u>	1.0471
24501	MAE	0.7126	0.7160	<u>0.7133</u>	0.7197

Table 3: Performance of DA-enhanced baselines (MSE/MAE), with the best result from Table 1 shown as a reference.

Data-set	Metr.	MM-T. (iTr.) DA	MM-T. (iTr.)	MM-T. (Pat.) DA	MM-T. (Pat.)	NLin. _DA	NLin.	MaTr. _DA	MaTr.	Uni. _DA	Uni.	T-LLM DA	T-LLM	Best in Tab.1
TMMD	MSE	1.3095↓	1.2430	1.1993↓	1.1842	1.2483↑	1.2869	1.1750↑	1.2000	1.8822↑	1.9075	2.2902↓	2.2487	1.0829
-H	MAE	0.6694↓	0.6653	0.6964↑	0.7005	0.7582↑	0.8114	0.7145↑	0.7234	0.8900↑	0.9025	1.1089↓	1.0309	0.6653
TMMD	MSE	0.4445↓	0.4299	0.4189↑	0.4513	0.4006↑	0.4854	1.0370↑	1.1749	1.2982↓	1.0529	0.5521↓	0.5517	0.4299
-A	MAE	0.4576↑	0.4718	0.4527↑	0.4900	0.4323↑	0.4686	0.7338↑	0.7643	0.8588↓	0.7848	0.5761↑	0.5896	0.4554
TMMD	MSE	0.6652↑	1.1604	1.1264↓	<u>1.0239</u>	1.1180↓	1.0645	1.1239↑	1.2241	1.1145↑	1.2081	1.2198↓	1.1798	1.0053
-C	MAE	0.9016↓	0.8984	0.8795↓	<u>0.8329</u>	0.8473↓	0.8239	0.8492↑	0.8815	0.8340↑	0.8782	0.8778↓	0.8663	0.8096
TMMD	MSE	0.0140↑	<u>0.0164</u>	0.0173↑	0.0181	0.0401↑	0.0448	0.7890↑	1.1295	3.4598↑	5.9978	0.0473↓	0.0406	0.0145
-Ec	MAE	0.0933↑	<u>0.1035</u>	0.1056↑	0.1086	0.1658↑	0.1685	0.8422↑	1.0094	1.6999↑	2.4338	0.1774↓	0.1655	0.0952
TMMD	MSE	0.1744↑	0.1825	0.1643↑	<u>0.1644</u>	0.1494↑	0.1654	0.2071↑	0.2985	0.7198↑	1.0659	0.3495↓	0.3272	0.1588
-En	MAE	0.2959↑	0.3049	0.2880↑	<u>0.2943</u>	0.2788↑	0.2979	0.3421↑	0.4101	0.6427↑	0.8020	0.4526↓	0.4401	0.2898
TMMD	MSE	0.4101↓	0.4042	0.4308↓	0.4124	0.4227↓	0.4117	0.3978↑	0.4310	0.5072↑	0.5555	0.4605↑	0.4635	0.3988
-Env	MAE	0.4726↓	0.4646	0.4672↓	<u>0.4621</u>	0.4680↑	0.4685	0.4638↑	0.4876	0.5493↑	0.5915	0.4880↓	0.4823	0.4612
TMMD	MSE	100.0897↑	101.6838	99.6815↑	100.6717	93.0014↑	<u>94.9721</u>	113.1712↓	112.9370	114.6106↑	114.6854	103.6998↓	95.1019	94.9721
-S	MAE	5.2849↓	5.2649	<u>5.2214↓</u>	5.1875	4.4651↑	<u>4.5411</u>	6.3443↓	6.3201	6.4487↑	6.4580	5.4526↓	4.8132	4.5411
TMMD	MSE	1.4173↓	1.2973	1.3202↑	1.4221	1.3917↓	1.3880	1.0070↓	0.8662	1.5128↑	1.7966	1.9071↑	1.9108	0.8662
-SG	MAE	0.6702↓	0.5762	0.6503↑	0.7207	0.7101↑	0.8353	0.5838↑	0.6282	0.8773↑	1.1014	1.0982↑	1.1052	0.5762
TMMD	MSE	0.2556↓	0.2291	0.2091↓	<u>0.2036</u>	0.2923↓	0.2651	0.2849↑	0.2857	0.6353↑	1.4763	0.3853↓	0.3720	0.1988
-T	MAE	0.2764↓	0.2524	0.2390↓	<u>0.2321</u>	0.3799↓	0.3754	0.4308↑	0.4331	0.6542↑	1.1122	0.4413↓	0.4352	0.2321
NF	MSE	0.0729↑	0.1118	0.0670↑	0.1134	0.0701↑	0.1326	0.8228↑	2.1544	2.0969↑	3.6883	0.1496↑	0.1498	0.0727
-FA	MAE	0.1140↑	0.2198	0.0998↑	0.2226	0.1260↑	0.2537	0.5824↑	1.1270	0.9859↑	1.5509	0.2129↑	0.2151	0.1086
NF	MSE	0.6561↓	0.4766	0.6092↑	1.1376	0.5150↑	1.0733	0.4389↓	<u>0.4196</u>	0.4461↑	0.6785	0.5843↑	0.5868	0.5681
-FC	MAE	0.5532↑	0.5562	0.5433↑	0.5521	0.5316↑	0.5355	0.5574↑	0.5702	0.6060↓	0.6054	0.5284↑	0.5275	0.5355
NF-T	MSE	1.2444↓	1.2313	1.2294↓	1.2250	1.2253↑	1.2301	0.5751↑	0.6257	1.2749↑	1.3301	1.3415↑	1.3427	0.6257
-MAE	MAE	0.7810↓	0.7787	0.7763↓	<u>0.7738</u>	0.7837↑	0.7914	0.4112↑	<u>0.4410</u>	0.7979↑	0.8196	0.8311↑	0.8299	0.4410
TCAP	MSE	0.8369↓	0.8112	0.8060↓	<u>0.7964</u>	0.7960↓	0.7793	0.7980↑	0.8941	0.9339↑	0.9696	0.8197↑	0.8159	0.7793
-WH	MAE	0.6732↓	0.6630	0.6609↓	<u>0.6571</u>	0.6559↓	0.6488	0.6457↑	0.7083	0.7424↑	0.7696	0.6688↓	0.6672	0.6488
TCAP	MSE	0.7556↓	0.7508	0.7376↓	<u>0.7363</u>	0.7378↓	0.7289	0.7946↑	0.8871	0.8966↑	0.9759	0.7649↓	0.7615	0.7289
-WN	MAE	0.6679↓	0.6659	0.6618↓	<u>0.6612</u>	0.6605↑	<u>0.6605</u>	0.6934↑	0.7442	0.7491↑	0.7982	0.6805↓	0.6786	0.6605
TCAP	MSE	0.7918↓	0.8006	0.8060↓	<u>0.8014</u>	0.7970↑	0.8084	0.9592↑	1.1199	1.1272↑	1.3613	0.8623↑	0.8601	0.7884
-WS	MAE	0.6546↓	<u>0.6542</u>	0.6590↓	0.6522	0.6567↓	0.6562	0.6817↑	0.7588	0.7771↑	0.8638	0.6804↑	0.6792	0.6522
TCAP	MSE	0.0900↑	0.1579	0.0937↑	0.1453	0.0922↑	0.1567	1.8272↑	2.6243	2.6846↑	4.4568	0.2154↓	0.2087	0.1008
-FM	MAE	0.2328↑	0.2938	<u>0.2237↑</u>	0.2826	<u>0.2234↑</u>	0.2892	1.0387↑	1.1054	1.0304↑	1.4979	0.3433↓	0.3349	0.2342
TCAP	MSE	0.4543↓	<u>0.4226</u>	0.4242↓	0.4240	0.4580↑	0.5744	0.7050↑	0.7633	0.6116↓	0.5743	0.6635↑	0.6675	0.3887
-HM	MAE	0.4857↓	<u>0.4622</u>	0.4548↑	0.4680	0.5172↑	0.5797	0.7048↑	0.7639	0.5330↑	0.5792	0.5840↑	0.6040	0.4573
TCAP	MSE	3.6938↑	3.7627	3.4469↓	<u>3.4236</u>	3.8359↑	4.0059	5.7147↑	6.5429	6.4403↓	6.3504	4.4794↑	4.3463	3.4236
-HP	MAE	0.8539↑	0.8935	0.8444↓	0.8391	0.8900↑	0.9988	1.3323↑	1.3960	1.4555↓	1.4169	1.0177↓	1.0041	0.8391
TTC-C	MSE	0.6568↓	0.6483	0.6372↓	0.6291	0.6420↓	0.6359	0.5957↓	0.5901	0.7403↑	0.7735	0.6732↑	0.6656	0.5901
-MAE	MAE	0.5791↑	0.5801	0.5722↓	0.5669	0.5763↑	0.5844	0.5713↓	0.5642	0.6496↑	0.6758	0.5944↓	0.5914	0.5642
TTC-M	MSE	1.6796↑	2.0028	1.3059↑	1.6166	1.1456↑	2.0776	3.7963↑	4.7009	4.8369↓	4.8152	2.9614↑	2.9943	1.5711
7679	MAE	0.8913↑	0.9633	0.7815↑	0.8590	0.7531↑	0.9612	1.3349↑	1.5874	1.7394↑	1.7428	1.1198↑	1.1264	0.8590
TTC-M	MSE	3.4697↓	3.4278	3.2362↓	3.1111	2.8149↑	3.1214	3.7031↓	3.2655	3.0296↑	3.2098	3.4765↓	3.3953	3.0221
8355	MAE	1.4331↓	1.4224	1.3593↓	1.3358	1.2760↑	1.3374	1.4835↓	1.3630	<u>1.2891↑</u>	<u>1.3245</u>	1.4158↓	1.3967	1.2955
TTC-M	MSE	1.5204↑	1.6077	1.4684↑	1.5335	1.2960↑	1.6863	2.7163↑	2.9719	3.6384↑	3.8612	1.6962↓	1.7278	1.4930
15432	MAE	0.9046↑	0.9446	0.8429↑	0.9203	<u>0.8279↑</u>	1.0290	1.2725↑	1.3949	1.5331↑	1.5907	1.0140↑	1.0202	0.8503
TTC-M	MSE	4.1661↑	4.1907	4.0806↓	3.9465	3.7047↑	4.0105	4.3156↑	4.6530	4.2608↑	4.4719	4.0463↑	4.0384	3.8226
20415	MAE	1.1908↑	1.1986	1.1723↓	<u>1.1328</u>	1.1079↑	1.1578	1.2803↑	1.3674	1.3527↑	1.4090	1.1939↑	1.1906	1.1328
TTC-M	MSE	1.3523↓	1.2793	1.3702↑	1.3914	0.9298↑	<u>1.0250</u>	1.5697↑	1.6490	2.2317↑	2.5287	1.6263↑	1.6894	1.0250
24501	MAE	0.7882↓	0.7746	0.8213↑	0.8426	0.6208↑	<u>0.7127</u>	0.9753↑	1.0592	1.2346↑	1.3292	0.9184↑	0.9509	0.7126

Table 4: Transfer effectiveness metrics across forecasting horizons (H=12, 24)

Horizon	Metr.	MM-T. (iTrans.)	MM-T. (Patch.)	NLin.	MaTrans.	Uni.	Time-LLM
H=12	AWR	41.67%	45.83%	70.83%	79.17%	83.33%	33.33%
	PWR	58.33%	58.33%	87.50%	79.17%	91.67%	62.50%
	MRI	14.08%	14.00%	15.34%	16.85%	27.01%	1.55%
	MRD	-3.88%	-3.10%	-2.99%	-11.88%	-0.11%	-6.12%
H=24	AWR	42.11%	42.11%	36.84%	42.11%	26.32%	10.53%
	PWR	47.37%	47.37%	68.42%	57.89%	57.89%	31.58%
	MRI	11.21%	9.98%	11.31%	12.75%	4.08%	2.37%
	MRD	-8.01%	-4.42%	-6.17%	-12.42%	-24.25%	-6.04%